Digital Signal Intelligence

A New Information Substrate for Insurance
White paper

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Abstract:

Digital Signal Intelligence (DSI) defines a new, externally observable information substrate for insurance, enabling deterministic, autonomous risk assessment, exposure estimation, loss propensity inference, and pricing at scale.

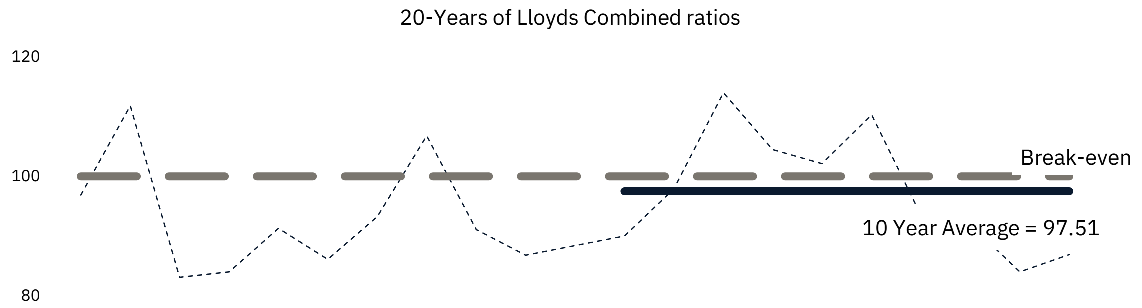
This paper presents a complete framework for transforming underwriting from subjective document interpretation to objective signal processing.

The Automation Myth

Introduction

For two decades, the insurance industry has attempted to automate the traditional underwriting process. Vendors, insuretechs, and internal transformation programmes have all pursued the same objective: extract structured data from unstructured submissions, feed it into rating models, and accelerate underwriting decisions.

Yet twenty years of Lloyd's combined ratios show the same truth as every major market: insurance is flat-lining. Despite billions invested, the results have been marginal. Zero-touch underwriting remains out-of-reach, expense ratios remain structurally high, and underwriting workflows remain dependent on manual intervention.



The core assumption behind these efforts is that traditional underwriting inputs can be automated.

This assumption is incorrect.

The inputs themselves are the constraint.

The Information Substrate Problem

Traditional inputs exist in infinite formats. Broker spreadsheets, PDFs, emails, and supplemental questionnaires have no canonical schema and no deterministic structure. No extraction system, including modern or future AI models, can consistently transform these artefacts into the reliable, field-level inputs required for zero-touch underwriting.

A system built on these inputs cannot achieve deterministic automation.

To enable autonomous underwriting, the information substrate must change.

Digital Signal Intelligence

The Core Thesis

Insurance underwriting requires understanding two properties of an organisation:

1. How well it is managed
2. How it behaves over time

Traditional underwriting attempts to infer these properties from self-reported documents, which are subjective, inconsistent, and non-verifiable.

DSI replaces them with observable digital signals.

All organisations operate within a digital ecosystem. Their participation in networks of clients, suppliers, partners, regulators, and certification bodies creates a citation network revealing trust and standing. Their digital trail of operational discipline provides a reliable proxy for maturity. Their hiring activity and investor communications reveal financial performance. Even the absence of expected signals is meaningful because well-run companies are predictable and uniform.

Signals are structured, verifiable, and machine-readable by design, which enables DSI to provide a deterministic foundation for underwriting, enabling automated risk assessment, exposure estimation, and pricing at scale.

Why Now?

DSI is not experimental. It is based on the same network-based principles that underpin modern search and recommendation systems, beginning with Google's PageRank in 1998. Just as PageRank introduced a static, graph-based substrate for ranking information, modern AI architectures extend this lineage through learned attention mechanisms capable of modelling complex relationships dynamically.

DSI applies this progression to organisational behaviour, treating digital signals as a structured network from which risk-relevant patterns can be inferred.

Today, the maturity of web, cloud technologies, and AI provides ideal conditions for success:

- Digital ubiquity.
  Every organisation of meaningful scale now has a measurable digital footprint, providing a universally present signal density.
- API accessibility.
  Signals that were locked in proprietary databases are now accessible through commercial and public APIs.
- Compute economics.
  Processing 200-400 signals per submission, running parallel extraction pipelines, and maintaining continuous monitoring was prohibitively expensive a decade ago. Cloud infrastructure has made this economical.
- AI maturation.
  Extraction, normalisation, and scoring of signals can now be executed by autonomous systems without human interpretation.

The question is not whether organisations can be assessed by their digital behaviour. The question is why insurance has taken so long to do so.

Foundational Principles

DSI operates consistently and autonomously.

To ensure this, ten mandatory principles define the methodology:

1. External Observability.
   Primary signals must be obtainable without cooperation.
2. Machine Readability.
   All signals must be analysable by automated systems without human interpretation.
3. Network Authority.
   Relationship patterns reveal quality. Models incorporate inbound authority, outbound selectivity, centrality, co-citation, and certification relationships.

4. Behavioural Inference Over Self-Reporting.
   Observable behaviour is weighted more heavily than stated intentions.
5. Absence as a Signal.
   Absence of expected signals is a measurable indicator of risk.
6. Structured Data Utilisation.
   Authoritative structured data—credit ratings, regulatory filings—are incorporated directly.
7. Minimal Direct Inquiry.
   Direct questions are optional and constrained to 5–10 binary or categorical questions per model.
8. Organisational-Level Assessment.
   Risk is assessed at the organisational level; parent behaviour is a proxy for underlying assets.
9. Simplicity in Scoring.
   Models follow a clear scoring flow that is captured and retained at the atomic grain.
10. Agentic Readiness.
    Models must be executable by autonomous systems.

Signal Architecture

Canonical Categories

DSI signals fall into seven canonical categories. All models must identify which categories they employ and document the specific signals within each.

The seven categories are:

1. Network Authority Signals.
   Signals derived from an entity's position within networks of trust, partnership, and transaction.
2. Technical Infrastructure Signals.
   Signals derived from observable technical implementation (TLS, DNS, exposed services, software currency, hosting infrastructure).
3. Corporate Digital Footprint Signals.
   Signals derived from an entity's own digital presence (website structure, content recency, policy documentation, leadership visibility, hiring activity).
4. Behavioural Signals.
   Signals derived from observable operational behaviour over time (change patterns, incident response, regulatory engagement, customer/employee interaction).
5. Public Record Signals.
   Signals derived from regulatory filings, litigation records, corporate registries, IP filings, and breach disclosures.
6. Structured Data Feed Signals.
   Signals derived from authoritative third-party providers (credit ratings, ESG scores, market data, classification society records).
7. Direct Inquiry Signals (Optional).
   Minimal, structured questions used only when externally observable signals are insufficient.

Absence as a Signal

Absence as a Signal is a foundational principle of DSI.

Where traditional underwriting treats missing data as unknown, requiring manual intervention to obtain it, DSI inverts this logic: absence is itself a signal. For example, well-run organisations have predictable digital footprints. A financial institution without security headers, a technology company without a careers page, a regulated entity without certification documentation—these absences are measurable negative indicators.

Absence signals are scored, not ignored. This enables DSI to differentiate between entities where traditional features (industry code, size, geography) appear identical.

Confidence as a First-Class Metric

Every signal carries a confidence score alongside its value. This dual-output design enables the system to know when it knows—and critically, when it does not.

Confidence scores (0–1) reflect three dimensions:

1. Data Availability: Was the signal successfully extracted from its source?
2. Source Reliability: How authoritative is the data source?
3. Temporal Decay: How recently was the data captured?

Weighted confidence propagates through composite scoring, producing an overall model confidence that determines decisioning authority.

End-to-End Model Architecture

Models convert raw signals into deterministic underwriting outputs through a structured workflow comprising six phases and fourteen discrete steps.

The structured workflow is:

Phase A – Discovery (Step 0)

- Entity identification from minimal viable input (typically client name and / or domain) triggers website discovery, validation, and corporate identity resolution.
- Outputs the discovered website URL, domain, confidence score, and corporate identity.

Phase B – Model Instantiation (Step 1)

- Configuration instantiation using content-addressable storage.
- Model version creation with full audit trail.
- Input verification against minimum viable requirements.

Phase C – Signals (Steps 2–4)

- Parallel signal extraction across all configured pipelines.
- Signals are normalised to a value between 0–100 using defined scoring algorithms. A confidence score (0–1) reflects availability, source reliability, and data freshness.

Phase D – Three Layer Assessment (Steps 5–7) – detailed in following section.

- Risk Layer.
  Pure composite score, on a 0-1000 scale, calculated.
  Composite scores map to configurable risk tier.
  Signal condition evaluation for tier overrides, referrals, modifiers, and notes.
- Exposure Shadow Layer.
  Calculate magnitude and complexity scores.
- Loss Correlation Layer.
  Calculates loss propensity and severity scores.

Phase E – Pricing (Steps 8–12)

- Final tier resolution applying maximum override rule. Each carries a pre-defined decision logic. For example, composite score ≥ 800 | auto-approve | excellent risk.
- Three Layer Assessment determines a base-rate premium.
- Deterministic modifiers are applied in sequence.
- Limit band scaling using ILF tables.

Phase F – Decision (Step 13)

- Final decision determination: Approve, Refer, or Decline.
- Full audit trail persistence.
- Model version finalisation.

Three Layer Assessment

Traditional underwriting conflates three distinct analytical assessments into a single, subjective, aggregated assessment. These are:

Layer	Assessment	Traditional Output
Risk	How well does this organisation manage risk?	Subjective Assessment.
Exposure	How large is the insurable exposure?	Self-reported SOV.
Loss	How likely is a loss event?	Historical Loss Data.

DSI separates these into independent, measurable analytical dimensions.

Using a Multi-Head Attention architecture, DSI independently assesses each dimension provided via the same underlying signal infrastructure, to determine their contextual relevance to a combined and unified pricing decision. Critically, each dimension is captured at the atomic grain.

Risk Assessment

Traditional risk assessment is built on inputs with no canonical schema and no deterministic structure. Underwriters interpret these artefacts differently, producing inconsistent decisions and preventing automation. Because the inputs are subjective, the outputs can never be calibrated.

DSI replaces this with a deterministic assessment based entirely on observable digital signals.

Organisational signals are assessed across the seven canonical categories, weighted and aggregated into a Composite Risk Score. This maps deterministically to a Risk Tier (typically, 1–5) which carries predefined decision logic. For example, Tier 1 | auto-approve.

The Risk Layer outputs six independent, risk focused dimensions:

1. Composite Risk Score.
   Weighted, aggregated score between 0-1000.
2. Risk Tier.
   Configurable categorisation, typically between 1-5.
3. Confidence Score.
   Confidence score between 0-1 reflects data availability, source reliability, and signal freshness.

4. Tier Overrides (if triggered).
   Deterministic rules that adjust the assigned tier when specific signal conditions are met (e.g., regulatory sanctions, critical security failures, governance vacancies).
5. Referral Flags.
   Explicit indicators that the risk requires human review due to low confidence, conflicting signals, or triggered conditions.
6. Deterministic Modifiers.
   Pre-defined, rule-based adjustments applied after tier assignment (e.g., industry-specific loadings, concentration adjustments). No subjective judgement is permitted.

Exposure Assessment

The lack of canonical schema and deterministic structure used for traditional exposure input prevents straight-through processing.

However, organisations leave measurable digital signal traces of their scale: the breadth of their digital footprint, the depth of their infrastructure, the number of locations implied by regulatory filings, the size of their partner and vendor networks, the complexity of their technology stack, and the geographic dispersion of their operations. These signals correlate strongly with exposure magnitude.

Using the same underlying signal architecture DSI correlates these to exposure magnitude using a tiered proxy hierarchy:

Tier	Category	Detail
1	Direct Observables	Market capitalisation, regulated asset disclosures, public financial filings. These provide the highest confidence estimates
2	High-Confidence Proxies	Employee count, office locations, regulatory jurisdiction count. Strong correlation with exposure, widely available
3	Inferred Signals	Subdomain complexity, technology stack depth, web traffic estimates. Lower confidence but high availability

The Exposure Shadow Layer outputs two independent, exposure focused dimensions:

1. Exposure Band.
   Classification of exposure magnitude.
2. Complexity Category.
   Assessment of exposure structure complexity based on geographic dispersion, subsidiary count, technology heterogeneity, and regulatory footprint.

Critically, DSI outputs ranges with confidence rather than false-precision point estimates. For example, a typical output might read: "Estimated TIV: $75M-$180M at 78% confidence, placing this risk in the LARGE exposure band." This honest uncertainty enables automated decisioning while flagging cases requiring verification.

Loss Assessment

Traditional loss analysis faces an insurmountable data problem. Loss events are episodic: 80-90% of policies have zero claims in any year. With sparse outcomes and minimal features, industry code, size, geography, causal attribution is nearly impossible. For example, two banks with $10 billion in assets can have radically different loss profiles based on factors invisible to traditional underwriting. When losses occur, the available features are insufficient to explain why. When losses don't occur, there's no data to learn from. This fundamental constraint ensures that the gap between loss-data and technical pricing remains absolute.

DSI inverts this limitation inferring loss propensity from the same observable signals used to assess risk and exposure. More importantly, these signals are leading indicators: they deteriorate before losses occur.

Loss events are preceded by measurable behavioural and structural patterns. DSI captures these patterns through signals that describe operational discipline, governance stability, security posture, regulatory engagement, partner and vendor quality, digital footprint vitality, and organisational change velocity.

The causal mechanism is straightforward:

- Weakening security posture (outdated TLS, unpatched systems) precedes cyber breaches.
- Declining governance transparency (reduced board disclosures, leadership turnover) precedes D&O actions.
- Regulatory friction (enforcement letters, compliance citations) precedes enforcement actions.
- Management instability (C-suite departures, rapid hiring, organisational restructuring) precedes operational failures.

These patterns are detectable because DSI observes behaviour continuously, not just at bind. The traditional insurer learns about risk quality at claim time. DSI enables learning before the loss occurs and can trigger proactive risk management engagement, renewal repricing, or non-renewal decisions before a loss occurs.

The Loss Correlation Layer outputs four independent, loss focused dimensions:

1. Loss Propensity Score (0-100).
   Likelihood of loss frequency based on signal patterns.
2. Severity Propensity Score (0-100).
   Likelihood of higher-severity outcomes.
3. Loss Trend Band.
   Trend classification of propensity over time from very low to high.
4. Cohort Assignment.
   Signal-derived peer groups that share loss characteristics.
   Cohorts enable several capabilities: peer comparison for outlier detection, cohort-level loss expectancy for pricing, and migration tracking as entities move between cohorts over time.

Worked Example

The following example demonstrates DSI's end-to-end assessment of a mid-market technology company seeking cyber coverage.

Submission: "TechFlow Solutions" with domain hint techflow.io

Phase A – Discovery (Step 0)

- DSI validates techflow.io as the corporate website (95% confidence) and identifies TechFlow Solutions Inc. as the legal entity through corporate registry matching.

Phase B – Model Instantiation (Step 1)

- Model version created, inputs verified against minimum viable requirements.

Phase C – Signals (Steps 2–4)

- Parallel extraction across 35 cyber-specific signals.

Phase D – Three Layer Assessment (Steps 5–7)

- Risk Layer.
  Composite Score: 782/1000 → Tier 2 (Good Risk)
  Confidence: 0.87 (High)
  Rationale: Good Technical Infrastructure, Positive Network Authority, Good Behavioural Signals, Corporate Footprint extensive however, 12 open positions including CISO role.
- Exposure Shadow Layer.
  Exposure Band: Medium ($10M-$50M implied TIV)
  Complexity Score: Moderate
  Exposure Confidence: 0.76
  Rationale: 38 subdomains, multi-cloud architecture, regulatory filings in 3 jurisdictions.
- Loss Correlation Layer.
  Loss Propensity: Low (score 28 / 100)
  Severity Propensity: Moderate (score 42 / 100)
  Loss Trend Band: Stable
  Cohort: “Tech Growth Mature Security”
  Rationale: governance stability, security posture trend, management tenure.

Phase E – Pricing (Steps 8–12)

- Final Premium Calculation ($5M limit): $20,250
  Base Rate: 0.45% of Limit
  Exposure Modifier: 1.00
  Complexity Modifier: 1.00
  Loss Modifier: 0.90
- No Modifiers Applied.

Phase F – Decision (Step 13)

- Decision: AUTO-APPROVE (Tier 2, confidence 0.87, no referral triggers)
- Total processing time: 47 seconds. Full audit trail persisted.

Competitive Differentiation

DSI occupies a fundamentally different position in the underwriting landscape than existing signal-based tools.

- Cyber Rating Services.
  Tools such as SecurityScorecard and BitSight provide security posture scores derived from technical infrastructure signals. These are valuable inputs to underwriting, but they are not underwriting systems. They do not assign risk tiers, calculate premiums, estimate exposure, infer loss propensity, or render decisions. An underwriter receiving a BitSight score must still interpret it, contextualise it, and manually integrate it into a pricing workflow.
- Other Coverages.
  No equivalent signal providers exist at scale. For example, within Energy no aggregation framework for OT/IT convergence indicators, safety culture signals, environmental compliance patterns, and operational discipline markers exists. DSI provides one.

This is a categorical distinction. DSI is not a better data feed—it is a complete underwriting substrate that happens to consume data feeds, including those from existing vendors, as inputs.

Economic Impact

The following projections are based on model architecture and assume signal availability ≥ 80% and straight-through processing rates ≥ 60%. Realisation timeline is estimated at 18–36 months from full deployment.

The insurance market currently operates with a structurally inefficient information substrate. The cost of that inefficiency is distributed across every participant in the chain, absorbed as expense ratio by capital providers, as processing cost by intermediaries, and as premium loading by the insured.

DSI does not redistribute this cost. It removes it.

Market-Level Efficiency Gains

Metric	Traditional Model	DSI Target
Time to quote	3–10 days	< 60 seconds
Cost per submission	$150–650	< $10
Straight-through rate	5–15%	60–80%
Model versioning	Manual, inconsistent	Automatic, immutable
Audit trail	Partial, reconstructed	Complete, immutable
Scalability	Linear (headcount)	Logarithmic (compute)

These efficiencies are not tied to any single participant's operating model. They are properties of the information substrate itself. Any market participant operating on DSI signals, whether a carrier, syndicate, MGA, or broker, realises the same structural efficiency improvement.

Expense Ratios

The current market expense ratio reflects the cost of operating on an unstructured, self-reported information substrate. DSI targets a structural reduction of approximately 22 points across the participant that deploys it:

Metric	Traditional Model	DSI Target
Staff Costs	12%	4%
Office and Infrastructure	8%	2%
Administration	4%	1%
Brokerage and Commissions	11%	6%
Total	35%	13%

The distribution of these gains across the market, between capital providers, intermediaries, and insureds, will be determined by competitive dynamics and adoption, not the methodology.

Loss Ratios

DSI targets a structural improvement in loss ratios through two mechanisms:

Source	Mechanism	DSI Target
Loss Signal Correlation	Behaviour-driven propensity	4 – 8 points
Continuous Calibration	Dynamic model tuning	2 – 4 points
Total		6 – 12 points

Overall Portfolio Impact

On a $500M premium book, the combined expense and loss ratio improvements represent $130M–$170M in improved underwriting profit. This figure is independent of which market participant captures it, the economic value exists in the methodology and accrues to whoever deploys it first and most completely.

Key Assumptions

Assumption	Value	Sensitivity
Signal availability rate	≥ 80%	±3 points per 10% variance
Straight-through processing	≥ 60%	±2 points per 10% variance
Model calibration period	12 months	N/A
Loss correlation validation	24–36 months	Deferred benefit

All projections are subject to portfolio composition and market conditions.

The Path to Market Standard

The traditional underwriting model has existed for over 300 years and has never produced an accepted assessment standard. This is not a historical accident; it is a structural impossibility. A model built on subjective judgment, that is neither canonical nor deterministic, cannot become a standard regardless of how long it operates.

DSI has the opposite characteristics. It is observable, every signal is externally verifiable. It is repeatable, the same entity scored twice produces the same result. It is consistent, the algorithm does not vary by individual or institution. It is auditable, every pricing decision traces through a defined signal-to-tier-to-premium chain. It is non-gameable, signals are drawn from behaviour, not self-reporting.

These are precisely the characteristics that allowed FICO to become the global credit standard, that allowed PageRank to define information quality, that allow rating agency outputs to function as regulatory reference points. The traditional underwriting model never had them. DSI does.

The adoption sequence follows a predictable path. A 12–24-month forward validation period — DSI-scored risks tracked against actual loss outcomes across early-adopting participants — produces the independent evidence base that transitions DSI from methodology to standard. At that point the dynamic accelerates without further persuasion: capital that has historically been excluded from insurance by its opacity enters specifically to back transparently scored risk, because the information asymmetry that made the asset class illegible has been removed. Early participants in that validation cohort do not merely gain a better tool — they become the institutions whose loss experience defines what each DSI tier means in practice.

Design Boundaries

DSI is designed for broad applicability but operates within defined boundaries:

- Digital Footprint Dependency.
  DSI requires entities to have a measurable digital presence. Risks with minimal digital footprint—small local businesses, individual artisans, entities operating primarily offline—may not generate

sufficient signal density for automated assessment. These risks are better suited to traditional underwriting approaches.

- Calibration Requirements.
  Initial model weights are based on actuarial judgment and industry analysis. Full loss-correlation calibration requires 24-36 months of loss emergence data. The continuous tuning architecture enables rapid recalibration as loss data accumulates.
- Catastrophe and Systemic Risk.
  DSI assesses individual entity quality, not correlated exposure. Natural catastrophe, pandemic, and systemic financial risk require separate accumulation management. The Exposure Shadow Layer provides accumulation proxies, but catastrophe modelling remains a distinct discipline.
- Bespoke and Unique Risks.
  Highly customised programmes—one-off construction projects, unique assets, novel business models—may lack comparable signal patterns. Referral thresholds ensure novel risks receive human review.

Conclusion

Digital Signal Intelligence replaces the traditional, self-reported information substrate of insurance with externally observable, machine-readable signals. This enables:

- Deterministic scoring without subjective interpretation
- Autonomous underwriting with deterministic guardrails at scale
- Exposure estimation without self-reported schedules
- Loss pattern inference from behavioural signals
- Continuous monitoring and early deterioration detection
- Global scalability without linear headcount growth

The insurance market has sought a viable path to these outcomes for two decades. The bottleneck has never been computing power, data availability, or ambition. It has been the information substrate itself — an unstructured, self-reported foundation that is structurally incapable of supporting deterministic automation regardless of how much is invested in processing it.

DSI resolves this at the foundation. The economic value it creates is real, it is measurable, and it is available to any market participant willing to move first.

DSI is a structural replacement for the information substrate of underwriting.

The question of who captures the value it creates is a competitive one. The question of whether that value exists is not.

As digital signals expand and calibration datasets grow, DSI becomes not only a new methodology for underwriting, but the foundation upon which the next generation of insurance market infrastructure, built for causal simulation and real-time World Model risk prediction, will be constructed.

Appendix

Key Terminology

- Coverage.
  A coverage is a defined insurance product (e.g., Cyber, D&O, Marine Hull). Each coverage contains one or more configurations.
- Configuration.
  A configuration is the complete specification of a model for a given coverage. It defines: the signals used, the scoring algorithms, the weightings, the tier thresholds, the pricing logic, the decision rules. A configuration is the single source of truth for how a model operates.
- Model Cycle.
  A model cycle is a complete execution of a configuration against a submission. Each cycle produces: a full signal extraction, a composite score, a risk tier, (optionally) an exposure band and complexity category, a pricing output, a decision. Every cycle is persisted as a Model Version, ensuring full auditability.
- Signal.
  A measurable, externally observable data point used in scoring. Signals are normalised to a 0–100 scale.
- Composite Score.
  A weighted aggregation of signals normalised to a 0–1000 scale. Composite scores map directly to risk tiers.
- Risk Tier.
  A discrete classification (typically Tier 1–5) representing risk quality. Tiers drive pricing and decision logic.
- Exposure Band.
  A classification of exposure magnitude (micro → very large), inferred from observable signals.
- Complexity Category.
  A classification of exposure complexity (simple → highly complex), inferred from observable signals.
- Deterministic Adjustments.
  Explicit, rule-based modifiers applied after tier assignment, defined in the configuration. No subjective adjustments are permitted.